Extremal Nonhamiltonian-Connected Graphs with Given Order and Connectivity *

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Abstract

Hamiltonian-connectedness is a strong form of hamiltonicity, requiring a Hamilton path between every pair of distinct vertices. Note that deciding whether a graph is hamiltonian-connected is NP-complete. Thus, from an extremal point of view, it is natural to study sufficient edge conditions for hamiltonian-connectedness by asking how many edges a graph can have while still failing to be hamiltonian-connected. Ore answered this question by determining the maximum number of edges in nonhamiltonian-connected graphs of a given order. Motivated by Ore's theorem, in this paper we determine the maximum number of edges in nonhamiltonian-connected graphs of order n and connectivity k, and characterize all extremal graphs. Our result may be viewed as the hamiltonian-connected analogue of the theorem of Zhan and Zhang on nonhamiltonian graphs with prescribed order and connectivity, and further refines Zhang's result on k-connected nonhamiltonian-connected graphs. An analogous result for nonpancyclic graphs is also presented.

Keywords: Hamiltonian-connected; Connectivity; Size

AMS Subject Classification: 05C38; 05C40; 05C45

1. Introduction

We consider finite simple graphs and use standard terminology and notation from [4] and [18]. Throughout, all graphs are assumed to be simple, finite and undirected. Let

*J. Yang is supported by National Key R&D Program of China under grant number 2024YFA1013900, NSFC under grant number 12471327. L. Zhang financially supported by JSPS KAKENHI Grant Number 25KF0036, the National Natural Science Foundation of China Grant Number 12601664, 12571365, the NSF of Hubei Province Grant Number 2025AFB309, the China Postdoctoral Science Foundation Grant Number 2025M773113, the Fundamental Research Funds for the Central Universities, Central China Normal University Grant Number CCNU24XJ026.

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$G = (V(G), E(G))$ be a graph with vertex set $V(G)$ and edge set $E(G)$. Then $n(G) := |V(G)|$ and $e(G) := |E(G)|$ are called the *order* and *size* of G, respectively. For $v \in V(G)$, let $N_G(v)$ and $d_G(v)$ be the neighborhood and the degree of v in G, respectively. We write $N_G[v] = N_G(v) \cup \{v\}$. Let $\delta(G) = \min\{d_G(v) : v \in V(G)\}$ be the minimum degree of G. The *connectivity* of a graph G, denoted by $\kappa(G)$, is the minimum number of vertices whose removal disconnects G or reduces it to a trivial graph. A graph G is said to be *k-connected* if $\kappa(G) \geq k$.

A Hamilton cycle (resp., path) in a graph G is a cycle (resp., path) containing every vertex of G. A graph G is called *hamiltonian* if it contains a Hamilton cycle. And G is called *hamiltonian-connected* if there exists a Hamilton path between every pair of distinct vertices of G. It is well known that the problems of deciding whether a graph is hamiltonian or hamiltonian-connected are NP-complete [10]. Consequently, a central theme in graph theory is to seek sufficient conditions for hamiltonicity. For related results, we refer the reader to [1, 11, 13, 17, 19, 23].

In 1961, Ore [14] determined the maximum number of edges in a nonhamiltonian graph of given order and characterized all extremal graphs. It is natural to consider analogous extremal problems for nonhamiltonian graphs under constraints on other graph parameters. Among these parameters, minimum degree and connectivity have been extensively studied. For example, Erdős [9] determined the maximum number of edges in an n-vertex nonhamiltonian graph with minimum degree at least k; see also [16]. Zhang [21] extended this result by determining the maximum number of edges in an n-vertex nonhamiltonian graph with minimum degree exactly k. Byer and Smeltzer [5] gave an upper bound on the number of edges in k-connected nonhamiltonian graphs. Recently, Zhan and Zhang [20] completely solved this problem and proved the following theorem.

Theorem 1 (Zhan-Zhang [20]). *Let $\phi(n, k)$ denote the maximum size of a nonhamiltonian graph of order n and connectivity k, where $n \geq 2k + 1$. Then*

$$\phi(n,k) = \begin{cases} \binom{n-k}{2} + k^2 & \text{if } n \text{ is odd and } n \geq 6k-5 \text{ or } n \text{ is even and } n \geq 6k-8, \\ \frac{3n^2-8n+5}{8} + k & \text{if } n \text{ is odd and } 2k+1 \leq n \leq 6k-7, \\ \frac{3n^2-10n+16}{8} + k & \text{if } n \text{ is even and } 2k+2 \leq n \leq 6k-10. \end{cases}$$

Hamiltonian-connectedness is a stronger property than hamiltonicity, and it is natural to consider the corresponding extremal problem for nonhamiltonian-connected graphs. One cornerstone in this direction is the following celebrated theorem of Ore [14]. To state it conveniently, we first introduce some notation. For two graphs G and H, $G \vee H$ denotes the *join* of G and H, that is, the graph obtained from the disjoint union $G + H$ by adding all edges joining a vertex of G to a vertex of H. Throughout the paper, equality of graphs is understood up to isomorphism; thus, $G = H$ means that G and H are isomorphic.

Theorem 2 (Ore [14]). *The maximum size of a nonhamiltonian-connected graph of order n is $\binom{n-1}{2} + 2$ and this size is attained by a graph G if and only if $G = K_3 \vee \overline{K_3}$ or $G = K_2 \vee (K_{n-3} + K_1)$.*

Note that the extremal graph has minimum degree 2 for all $n \geq 7$. Since hamiltonian-connected graphs have minimum degree at least 3, it is natural to consider similar graphs with a larger minimum degree. Zhou and Wang [25] proved that for $n \geq 13$, the maximum number of edges in a nonhamiltonian-connected graph of order n and minimum degree at least 3 is $\binom{n-2}{2} + 6$. Ho, Lin, Tan, Hsu, and Hsu [11] determined the maximum size of a nonhamiltonian-connected graph with a given order and minimum degree at least δ. Recently, Zhang [22] determined the maximum number of edges in a nonhamiltonian-connected graph with prescribed order and minimum degree.

Connectivity is one of the most fundamental measures of the robustness of a graph and plays an important role in hamiltonian graph theory. A classical cornerstone relating connectivity to hamiltonian-connectedness is the following theorem of Chvátal and Erdős.

Theorem 3 (Chvátal-Erdős [6]). *Let G be a nonhamiltonian-connected graph of connectivity $k \geq 2$. Then $\alpha(G) \geq k$, where $\alpha(G)$ denotes the independence number of G.*

Along this line of research, Zhou, Broersma, Wang and Lu [24] determined the maximum spectral radius of all k-connected nonhamiltonian-connected graphs of a given order. Zhang [22] recently determined the maximum number of edges in a k-connected nonhamiltonian-connected graph of prescribed order.

Theorem 4 (Zhang [22]). *Let G be a k-connected ($k \geq 3$) nonhamiltonian-connected graph of order n. Then*

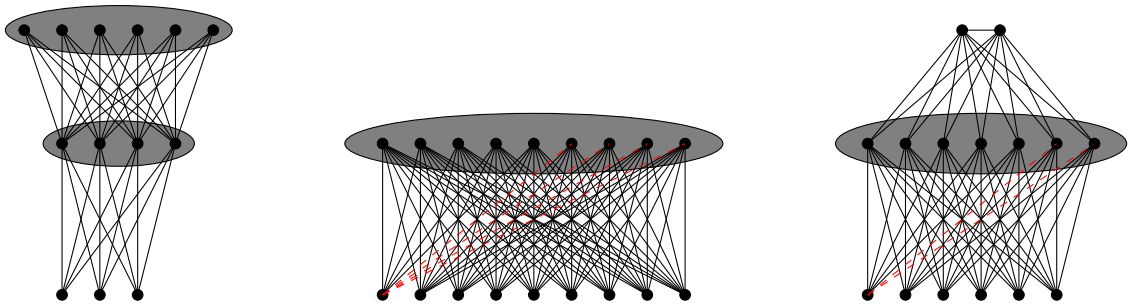
$$e(G) \leq \max\left\{\binom{n-k+1}{2} + k(k-1),\ \binom{n-\lfloor \frac{n}{2} \rfloor + 1}{2} + \left\lfloor \frac{n}{2} \right\rfloor \left(\left\lfloor \frac{n}{2} \right\rfloor - 1 \right)\right\}.$$

It is worth noting that, for some values of n and k, an extremal graph in Theorem 4 may have connectivity substantially larger than k. For instance, the maximum number of edges in a 4-connected nonhamiltonian-connected graph of order 16 is 92, but the extremal graph attaining this bound has connectivity 8, not 4. This naturally leads us to study the sharper problem of determining the maximum number of edges in a nonhamiltonian-connected graph with prescribed order and connectivity. To this end, we first introduce two families of graphs.

Definition 1. *Let n and k be positive integers. Define $F_{n,k} := K_k \vee \left(K_{n-2k+1} + \overline{K_{k-1}}\right)$; see Figure 1(a). Let $X_{n,k}$ denote the graph obtained from $K_{\lfloor \frac{n}{2} \rfloor} \vee \left(K_{\lceil \frac{n}{2} \rceil - \lfloor \frac{n}{2} \rfloor + 1} + \overline{K_{\lfloor \frac{n}{2} \rfloor - 1}}\right)$ by deleting $\lfloor \frac{n}{2} \rfloor - k$ edges that are incident to one common vertex in $\overline{K_{\lfloor \frac{n}{2} \rfloor - 1}}$; see Figure 1(b) and (c).* [*]

Dirac [8] showed that a graph G of order n is hamiltonian-connected if $\delta(G) \geq (n+1)/2$. In 1963, Ore [15] obtained that G is hamiltonian-connected if $d_G(x) + d_G(y) \geq n + 1$ for any pair of non-adjacent vertices $x, y \in V(G)$. Thus, for the existence of a nonhamiltonian-connected graph of order n and connectivity k we necessarily have $n \geq 2k$. Now we are ready to state and prove the main result.

[*]Note that $X_{n,k} = F_{n,k}$ if $n \in \{2k, 2k+1\}$.



(a) $F_{13,4}$ (b) $X_{18,5}$ (c) $X_{15,5}$

Figure 1: Three example graphs with deleted dashed edges, the vertices in each gray ellipse induce a complete graph

Theorem 5. *Let $f(n,k)$ denote the maximum size of a nonhamiltonian-connected graph of order $n \geq 3$ and connectivity $k \geq 2$, where $n \geq 2k$. Then*

$$f(n,k) = \begin{cases} \binom{n-k+1}{2} + k(k-1) & \text{if } n \text{ is odd and } n \geq 6k-13 \text{ or } n \text{ is even and } n \geq 6k-10, \\ \frac{3n^2-6n}{8} + k & \text{if } n \text{ is even and } 2k \leq n \leq 6k-12, \\ \frac{3n^2-8n+13}{8} + k & \text{if } n \text{ is odd and } 2k+1 \leq n \leq 6k-15. \end{cases}$$

Moreover, the extremal graphs are characterized as follows.

- *If $n \in \{6k-10, 6k-13\}$, then $f(n,k)$ is attained by a graph G if and only if $G \in \{F_{n,k}, X_{n,k}\}$.*
- *If n is odd and $n \geq 6k-11$, or if n is even and $n \geq 6k-8$, then $f(n,k)$ is attained by a graph G if and only if $G = F_{n,k}$.*
- *If n is even and $2k \leq n \leq 6k-12$, or if n is odd and $2k+1 \leq n \leq 6k-15$, then $f(n,k)$ is attained by a graph G if and only if $G = X_{n,k}$.*

In Section 2, we prove Theorem 5.

2. Proof of Theorem 5

To prove Theorem 5, we will need the following lemmas.

Lemma 6 (Lick [12])**.** *Let G be a graph with degree sequence $d_1 \leq d_2 \leq \cdots \leq d_n$, where $n \geq 3$. If $d_{i-1} \leq i$ implies that $d_{n-i} \geq n-i+1$ for $2 \leq i \leq n/2$, then G is hamiltonian-connected.*

Lemma 7 (Bondy [3])**.** *Let G be a graph of order n with degree sequence $d_1 \leq d_2 \leq \cdots \leq d_n$. Let r be an integer with $0 \leq r \leq n-2$. If for every integer j satisfying $1 \leq j \leq n-1-d_{n-r}$ we have $d_j \geq j+r$, then G is $(r+1)$-connected.*

Lemma 8. *Let n and k be two integers with $n \geq 2k$.*

$$\max\{e(F_{n,k}),\ e(X_{n,k})\}$$

$$= \begin{cases} \binom{n-k+1}{2} + k(k-1) & \text{if } n \text{ is odd and } n \geq 6k-13 \text{ or } n \text{ is even and } n \geq 6k-10, \\ \frac{3n^2-6n}{8} + k & \text{if } n \text{ is even and } 2k \leq n \leq 6k-12, \\ \frac{3n^2-8n+13}{8} + k & \text{if } n \text{ is odd and } 2k+1 \leq n \leq 6k-15. \end{cases}$$

Proof. By an elementary computation, we get

$$e(F_{n,k}) = \binom{n-k+1}{2} + k(k-1), \quad \text{and} \quad e(X_{n,k}) = \binom{\lceil \frac{n}{2} \rceil + 1}{2} + \left\lfloor \frac{n}{2} \right\rfloor \left(\left\lfloor \frac{n}{2} \right\rfloor - 2 \right) + k.$$

We first assume that n is odd. Then

$$e(X_{n,k}) = \frac{3n^2 - 8n + 13}{8} + k.$$

A direct calculation gives

$$e(F_{n,k}) - e(X_{n,k}) = \frac{(2k-n+1)(6k-n-13)}{8}.$$

Since n is odd and $n \geq 2k$, we have $n \geq 2k+1$. Thus $2k-n+1 \leq 0$, with equality if and only if $n = 2k+1$. For $n > 2k+1$, the sign is determined by $6k-n-13$. Therefore, $e(F_{n,k}) > e(X_{n,k})$ if $n > 6k-13$, $e(F_{n,k}) < e(X_{n,k})$ if $2k+1 < n < 6k-13$, and $e(F_{n,k}) = e(X_{n,k})$ if $n \in \{2k+1, 6k-13\}$.

Next we assume that n is even. Then

$$e(X_{n,k}) = \frac{3n^2 - 6n}{8} + k.$$

A direct calculation gives

$$e(F_{n,k}) - e(X_{n,k}) = \frac{(2k-n)(6k-n-10)}{8}.$$

Since $n \geq 2k$, we have $2k - n \leq 0$, with equality if and only if $n = 2k$. For $n > 2k$, the sign is determined by $6k - n - 10$. Therefore, $e(F_{n,k}) > e(X_{n,k})$ if $n > 6k-10$, $e(F_{n,k}) < e(X_{n,k})$ if $2k < n < 6k-10$, and $e(F_{n,k}) = e(X_{n,k})$ if $n \in \{2k, 6k-10\}$. This proves Lemma 8. □

A graph G is said to be *t-tough* if $t \cdot w(G-X) \leq |X|$ for all $X \subseteq V(G)$ with $w(G-X) \geq 2$, where t is a nonnegative real number and $w(G-X)$ is the number of components of $G-X$. The *toughness* $\tau(G)$ of the graph G is the maximum value of t such that G is t-tough (taking $\tau(K_n) = \infty$ for all $n \geq 1$, where K_n denotes the complete graph of order n). The concept of toughness of a graph was introduced by Chvátal [7]. Clearly, every hamiltonian-connected graph has toughness greater than one.

We are now ready to prove Theorem 5.

Proof of Theorem 5. Let $f(n,k)$ denote the maximum size of a nonhamiltonian-connected graph of order $n \geq 3$ and connectivity $k \geq 2$. Since the toughnesses of $F_{n,k}$ and $X_{n,k}$ are at most 1, they are nonhamiltonian-connected graphs of order n with connectivity k. Consequently, $f(n,k) \geq \max\{e(F_{n,k}), e(X_{n,k})\}$.

Next we prove $f(n,k) \le \max\{e(F_{n,k}), e(X_{n,k})\}$. Let G be a nonhamiltonian-connected graph of order n and connectivity k with degree sequence $d_1 \le d_2 \le \cdots \le d_n$. By Lemma 6, there exists an integer i such that

$$2 \le i \le \frac{n}{2}, \quad d_{i-1} \le i \quad \text{and} \quad d_{n-i} \le n-i.$$

Since $\delta(G) \ge \kappa(G) = k$, we have $i \ge d_{i-1} \ge \delta(G) \ge k$. Thus

$$k \le i \le \left\lfloor \frac{n}{2} \right\rfloor. \tag{2.1}$$

Based on $\kappa(G) = k$ and Lemma 7, we have that there exists an integer j such that

$$1 \le j \le n-1-d_{n-k} \quad \text{and} \quad d_j \le j+k-1. \tag{2.2}$$

Thus

$$d_{n-k} \le n-1-j. \tag{2.3}$$

Since $d_{i-1} \le i$, the first $i-1$ degrees are at most i. Since $d_{n-i} \le n-i$, the next $n-2i+1$ degrees are at most $n-i$. By (2.3), the next $i-k$ are at most $n-1-j$, and the last k are at most $n-1$. Therefore, the degrees of G are controlled by the following degree sequence.

$$\underbrace{i,\ldots,i}_{i-1},\ \underbrace{n-i,\ldots,n-i}_{n-2i+1},\ \underbrace{n-1-j,\ldots,n-1-j}_{i-k},\ \underbrace{n-1,\ldots,n-1}_{k},$$

If $j = 1$, then by (2.2), $d_1 \le k$. This together with $\delta(G) \ge k$ implies $d_1 = k$. Therefore

$$2e(G) = \sum_{\ell=1}^{n} d_\ell \le k + (i-2)i + (n-2i+1)(n-i) + (i-k)(n-2) + k(n-1). \tag{2.4}$$

If $j \ge 2$, then

$$2e(G) = \sum_{\ell=1}^{n} d_\ell \le (i-1)i + (n-2i+1)(n-i) + (i-k)(n-1-j) + k(n-1). \tag{2.5}$$

The difference of the right-hand side in (2.4) and (2.5) is $(i-k)(j-2)$, which is nonnegative by (2.1). Therefore,

$$\begin{aligned} 2e(G) &\le (n-2i+1)(n-i) + k(n-1) \\ &\quad + \max\{k + (i-2)i + (i-k)(n-2), (i-1)i + (i-k)(n-1-j)\} \\ &= (n-2i+1)(n-i) + k(n-1) + k + (i-2)i + (i-k)(n-2) \\ &= 3i^2 - (2n+5)i + n^2 + n + 2k. \end{aligned} \tag{2.6}$$

Let $g(i) := 3i^2 - (2n+5)i + n^2 + n + 2k$. This is a quadratic polynomial in i with positive leading coefficient. Therefore, on the interval (2.1), its maximum is attained at an endpoint. Hence

$$2e(G) \le \max\{g(k), g(\lfloor n/2 \rfloor)\}.$$

By an elementary computation, we get

$$\frac{g(k)}{2} = \binom{n-k+1}{2} + k(k-1) = e(F_{n,k}),$$

and

$$\frac{g(\lfloor n/2 \rfloor)}{2} = \binom{\lceil \frac{n}{2} \rceil + 1}{2} + \left\lfloor \frac{n}{2} \right\rfloor \left(\left\lfloor \frac{n}{2} \right\rfloor - 2 \right) + k = e(X_{n,k}).$$

Therefore

$$e(G) \leq \frac{1}{2} \max \{g(k), g(\lfloor n/2 \rfloor)\} = \max\{e(F_{n,k}), e(X_{n,k})\}.$$

Next we characterize the extremal graphs. Assume that G has size $\frac{1}{2}\max\{g(k), g(\lfloor n/2 \rfloor)\}$. Then every inequality above becomes equality.

Case 1. The maximum of $g(i)$ is attained at $i = k$.

Then the right-hand side in (2.4) is equal to the right-hand side in (2.5). Since the inequality in (2.6) becomes equality, the degree sequence of G is

$$\underbrace{k, \ldots, k}_{k-1}, \underbrace{n-k, \ldots, n-k}_{n-2k+1}, \underbrace{n-1, \ldots, n-1}_{k}.$$

Let W be the set of $k-1$ vertices of degree k, let Y be the set of $n-2k+1$ vertices of degree $n-k$, and let U be the remaining set of k vertices of degree $n-1$.

Every vertex of W is adjacent to each vertex of U. Since each vertex of W is exactly k, it has no other neighbors. Combining this with the fact that each vertex of Y has degree $n-k$, we have that $Y \cup U$ is a clique. Therefore, $G = K_k \vee \left(K_{n-2k+1} + \overline{K_{k-1}}\right) = F_{n,k}$, as desired.

Case 2. The maximum of $g(i)$ is attained at $i = \lfloor n/2 \rfloor$ and $j = 1$.

Since the inequality in (2.6) becomes equality, the degree sequence of G is

$$k, \underbrace{\left\lfloor \frac{n}{2} \right\rfloor, \ldots, \left\lfloor \frac{n}{2} \right\rfloor}_{\lfloor \frac{n}{2} \rfloor - 2}, \underbrace{\left\lceil \frac{n}{2} \right\rceil, \ldots, \left\lceil \frac{n}{2} \right\rceil}_{\lceil \frac{n}{2} \rceil - \lfloor \frac{n}{2} \rfloor + 1}, \underbrace{n-2, \ldots, n-2}_{\lfloor \frac{n}{2} \rfloor - k}, \underbrace{n-1, \ldots, n-1}_{k}.$$

Let x be the unique vertex of degree k, let R be the set of $\left\lfloor \frac{n}{2} \right\rfloor - 2$ vertices of degree $\left\lfloor \frac{n}{2} \right\rfloor$, let D be the set of $\left\lceil \frac{n}{2} \right\rceil - \left\lfloor \frac{n}{2} \right\rfloor + 1$ vertices of degree $\left\lceil \frac{n}{2} \right\rceil$, let W be the set of $\left\lfloor \frac{n}{2} \right\rfloor - k$ vertices of degree $n-2$, and let U be the set of k vertices of degree $n-1$.

Since every vertex of U is dominating and $d(x) = k$, we have $N(x) = U$. Note that each vertex $w \in W$ has degree $n-2$, and w is not adjacent to x. Then it is adjacent to every vertex in $V(G) \setminus \{x\}$. Consequently $U \cup W$ is a $\left\lfloor \frac{n}{2} \right\rfloor$-clique and is joined to every vertex of $D \cup R$.

Since every vertex of R already has exactly $\left\lfloor \frac{n}{2} \right\rfloor$ neighbors in $U \cup W$, we have that R is independent and every vertex of R is non-adjacent to any vertex of $D \cup \{x\}$. Since every vertex of D is adjacent to all vertices of $U \cup W$ and has degree $\left\lceil \frac{n}{2} \right\rceil$, we have that $G[D] = K_1$ if n is even, and $G[D] = K_2$ if n is odd. This proves that $G = X_{n,k}$, as desired.

Case 3. The maximum of $g(i)$ is attained at $i = \lfloor n/2 \rfloor$ and $j \geq 2$.

If $\left\lfloor \frac{n}{2} \right\rfloor = k$, then by (2.1), $i = k$ and this case has already been covered in Case 1. Hence assume $\left\lfloor \frac{n}{2} \right\rfloor \geq k+1$ and $i \geq k+1$. We claim that

$$j = 2. \tag{2.7}$$

Suppose to the contrary that $j \geq 3$. Then the difference of the right-hand side in (2.4) and (2.5) is $(i-k)(j-2) > 0$. Thus the maximum is attained at $j = 1$, which contradicts the assumption that $j \geq 2$. Hence (2.7) holds.

Since the inequality in (2.6) becomes equality, we have $d_2 = \left\lfloor \frac{n}{2} \right\rfloor$. Combining this with (2.7) and (2.2) gives

$$\left\lfloor \frac{n}{2} \right\rfloor = d_2 \leq 2 + k - 1 = k + 1.$$

Thus $\left\lfloor \frac{n}{2} \right\rfloor \leq k+1$. This together with the assumption that $\left\lfloor \frac{n}{2} \right\rfloor \geq k+1$ implies

$$\left\lfloor \frac{n}{2} \right\rfloor = k + 1. \tag{2.8}$$

Now we consider the following two subcases.

Subcase 3.1. n is even.

Then $n = 2\left\lfloor \frac{n}{2} \right\rfloor = 2k+2$. Thus the degree sequence of G is

$$\underbrace{k+1, \ldots, k+1}_{k+1},\ n-3,\ \underbrace{n-1, \ldots, n-1}_{k}.$$

Let U be the set of k vertices of degree $n-1$. Then $G = K_k \vee H$, where $H = G - U$ has degree sequence

$$\underbrace{1, \ldots, 1}_{k+1},\ k-1. \tag{2.9}$$

Since $\kappa(G) = k$, the graph H must be disconnected. Otherwise G is $(k+1)$-connected, a contradiction. The unique disconnected graph with degree sequence in (2.9) is $K_{1,k-1} + K_2$. Therefore,

$$G = K_k \vee (K_{1,k-1} + K_2).$$

For $k \geq 3$, it is straightforward to verify that G is Hamiltonian-connected, a contradiction. Next assume that $k = 2$. Then $n = 6$ and

$$e(G) = e(K_2 \vee (K_{1,1} + K_2)) = 11 < 12 = e(F_{6,2}),$$

a contradiction.

Subcase 3.2. n is odd.

Then by (2.8), we have

$$n = 2\left\lfloor \frac{n}{2} \right\rfloor + 1 = 2k + 3.$$

Thus the degree sequence of G is

$$\underbrace{k+1,\ldots,k+1}_{k},\ \underbrace{k+2,k+2}_{2},\ n-3,\ \underbrace{n-1,\ldots,n-1}_{k}.$$

Again let U be the set of k vertices of degree $n-1$. Then $G = K_k \vee H$, where $H = G - U$ has degree sequence

$$\underbrace{1,\ldots,1}_{k},\ \underbrace{2,2}_{2},\ k.$$

Similarly, since $\kappa(G) = k$, the graph H must be disconnected. Let z be the unique vertex of degree k in H. The component containing z has at least $k+1$ vertices. Since H has order $k+3$ and contains no isolated vertices, the other component has exactly two vertices, and hence is K_2. The component containing z consists of z, its k neighbors, two of which have degree 2 and the remaining $k-2$ of which have degree 1. The two vertices of degree 2 must be adjacent to each other. Therefore $H = (K_{1,k} + e) + K_2$. Consequently

$$G = K_k \vee ((K_{1,k} + e) + K_2).$$

For $k \geq 3$, one sees that G is hamiltonian-connected, a contradiction. If $k = 2$, then $n = 7$ and

$$e(G) = e(K_2 \vee ((K_{1,2} + e) + K_2)) = 15 < 17 = e(F_{7,2}),$$

a contradiction.

Thus Case 3 cannot happen if $e(G)$ attains the maximum. Therefore every extremal graph arises as follows: if the maximum is attained at $i = k$, then $G = F_{n,k}$; if it is attained at $i = \lfloor n/2 \rfloor$, then $G = X_{n,k}$. Combining this with Lemma 8 proves Theorem 5. □

3. The Pancyclicity

We now determine the maximum number of edges in nonpancyclic graphs with prescribed order and connectivity. This result may be viewed as a corollary of Theorem 1. Before stating this result, we first recall a well-known theorem of Bondy.

Theorem 9 (Bondy [2]). *If G is an n-vertex hamiltonian graph with $e(G) \geq n^2/4$, then either $G = K_{n/2,n/2}$ or G is pancyclic.*

As a corollary, Bondy obtained the following theorem.

Theorem 10 (Bondy [2]). *Let G be a graph of order n. If $d_G(u) + d_G(v) \geq n$ for every pair of nonadjacent vertices $u, v \in V(G)$, then either G is pancyclic or $G = K_{n/2,n/2}$.*

Let $p(n,k)$ denote the maximum size of a nonpancyclic graph of order n and connectivity k. Let G be a nonpancyclic graph of order n, connectivity k and size $e(G) = p(n,k)$. If $n \leq 2k$, then $\delta(G) \geq n/2$. Based on the fact that G is nonpancyclic and Theorem 10, we have $G = K_{n/2,n/2}$. Furthermore, if $n \leq 2k-1$, then $\kappa(G) = \kappa(K_{n/2,n/2}) = n/2 \neq k$, which contradicts the choice of G. Thus $n = 2k$, $p(n,k) = e(G) = e(K_{k,k}) = k^2$, and $K_{k,k}$ is the unique extremal graph. Next we determine $p(n,k)$ for $n \geq 2k+1$.

Corollary 11. *Let $\phi(n,k)$ be the function defined in Theorem 1. If $n \geq 2k+1$, then $p(n,k) = \phi(n,k)$.*

Proof of Corollary 11. On the one hand, the extremal graphs of Theorem 1 are nonhamiltonian, and hence nonpancyclic. Therefore,

$$p(n,k) \geq \phi(n,k). \tag{3.10}$$

On the other hand, let Q be a nonpancyclic graph of order n with connectivity k and size $e(Q) = p(n,k)$, where $n \geq 2k+1$. We claim that Q is nonhamiltonian. Suppose to the contrary that Q is hamiltonian. Since $n \geq 2k+1$, $\kappa(K_{n/2,n/2}) = n/2 \neq k$ and hence $Q \neq K_{n/2,n/2}$. Combining this with the fact that Q is nonpancyclic and Theorem 9, we have

$$e(Q) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

Moreover, for $n \geq 2k+1$, one can verify that

$$\left\lfloor \frac{n^2}{4} \right\rfloor < \phi(n,k).$$

Combining this with (3.10), in this case we have $e(Q) < \phi(n,k) \leq p(n,k)$, which contradicts the assumption of $e(Q) = p(n,k)$. This implies that Q is nonhamiltonian. Then by Theorem 1, $p(n,k) = e(Q) \leq \phi(n,k)$. This together with (3.10) implies that $p(n,k) = \phi(n,k)$ for $n \geq 2k+1$, and all extremal graphs are precisely the nonhamiltonian extremal graphs appearing in Theorem 1. □

Acknowledgement

The authors would like to express their sincere gratitude to Long-Tu Yuan for fruitful discussions.

Declaration

Declaration on the use of AI During the preparation of this work, the authors used OpenAI's ChatGPT (GPT-5.6 Pro) for language editing. The authors subsequently reviewed and revised the text as necessary, independently verified all mathematical arguments, and take full responsibility for the content of the article.

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability No data were used for the research described in this article.

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